Meeting Summarizer: AI Tool that Converts Meeting Transcripts into Action Plans *

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***Abstract*—In today's workplaces, meetings generate large amounts of unstructured conversational data, making it difficult for participants to recall decisions, track responsibilities, and follow up on tasks. Existing meeting summarization tools primarily provide transcripts or generic summaries, lacking structured insights, accountability, and clear action points. This often leads to missed outcomes and reduced organizational efficiency. This paper proposes an AI-powered Meeting Summarizer that transforms meeting transcripts into structured and actionable outputs. The system integrates Speech-to-Text (STT – Speech To Text) processing, Natural Language Processing (NLP – Natural Language Processing), and Transformer-based summarization to identify key discussions, decisions, and action items. The action extraction module automatically detects assigned individuals, responsibilities, and deadlines from the transcript. The proposed system supports multilingual inputs and follows a modular, privacy-conscious architecture suitable for real-time and post-meeting analysis. Experimental evaluation demonstrates strong performance, achieving approximately 90–95% transcription accuracy, 0.75–0.85 ROUGE (Recall-Oriented Understudy for Gisting Evaluation) scores for summarization, and 85–90% accuracy in action item extraction. By converting unstructured meeting data into clear and actionable plans, the system enhances clarity, accountability, and overall efficiency in collaborative environments.**

***Index Terms*—Meeting summarization, action item extraction, speech-to-text, natural language processing, transformer models, decision detection, meeting analytics, multilingual AI.**

I. Introduction

In recent years, virtual and hybrid work have been adopted. Within environments, there has been a sharp rise in the number and length of professional meetings. While meetings are essential, they make collaborative decision-making and coordination possible. In many cases produce substantial amounts of unstructured conversational data. Participants have problems with remembering important. Decision making, delegation of duties, and post-assessment. interviews, which contribute to poor communication, time wastage, and minimize productivity. The majority of available meeting assistance tools are mainly concerned with producing untidy transcripts or brief overviews of discussions. These solutions enhance accessibility and despite the fact that they enhance accessibility, accessibility is still lacking. record- keeping, they do not offer organised results. as action plans, ownership of tasks and deadlines. As a result, critical resolutions and tasks were deliberated over in meetings, and got lost in the tedious transcripts that need to be found. Further human input effort on interpretation and execution. In order to overcome this shortcoming, the proposed Meeting Summarizer introduces an artificial intelligence-based system that aims at transforming meet-coding into systematic and participatory action plans rather than simply making passive summaries.

The system uses speech-to-text processing and natural language processing methods. To identify major parts of the discussion, choices, and task-relevant statements during meetings. Through context and language analysis clues, the system extracts action items and attributes responsible. Participants, and finds pertinent timelines. Moreover, the suggested solution has a design that is aimed scaling, multilingual and modular architecture. meetings and real time and post meeting analysis. By focusing on clarity, accountability, and automation, the Meeting Summarizer aims to reduce cognitive load on participants and improve execution efficiency. The remainder of this paper discusses the related work, system design, implementation methodology, and evaluation strategy, demonstrating the effectiveness of AI-assisted meeting intelligence in modern collaborative environments.

II. Related Work

The growing use of virtual and hybrid meetings has led to a lot of research on automated meeting analysis and summarization systems. Current solutions help users by transcribing, summarizing, or analyzing meeting conversations, but their effectiveness and coverage differ quite a bit. Generally, previous work in this area falls into three categories: (A) transcript-based meeting documentation systems, (B) summarization-focused meeting analysis tools, and (C) machine learning-based action and decision extraction systems. This section looks at these methods and points out the limitations that drive the need for the proposed Meeting Summarizer..

A. Transcript-Based Meeting Documentation Systems

Early meeting analysis systems mainly focused on automatic speech recognition to turn meeting audio into text transcripts. Commercial tools, like meeting recorders and conferencing platforms, create verbatim transcripts to improve accessibility and documentation. Although these systems increase information availability, they depend on users to manually review transcripts to find key points, decisions, and tasks.

Studies have shown that long conversation transcripts create a high cognitive load and are ineffective for recalling and executing tasks after meetings [1]. Furthermore, transcription-only approaches do not provide structured outputs or accountability, limiting their practical usefulness in professional environments.

B. Meeting Summarization Approaches

New studies have taken a different direction of automatic meeting summarization based on extractive and abstractive methods. Extractive methods are salient utterances that are chosen according to statistical. or linguistic characteristics, abstractive methods make use of. transformer-based models to come up with short summaries. Some studies prove that abstractive summarization is easier to read and comprehend than extractive. summaries [2], [3]. Nevertheless, the majority of summary systems. Create general or narrative summaries which are not explicit. assignment of decision, duties, and timelines. As a consequence, summaries usually are descriptive more than action-oriented, necessitating some additional human interpretation.

C. Machine Learning-Based Action and Decision Extraction

More advanced approaches attempt to extract action items, decisions and intents of meeting discussions based on supervised and semi-supervised machine learning models. Past studies have investigated classification of dialogue act, intent detection and task extraction to enhance meeting intelligence [7] , [10]. Despite their potential, these methods tend to be difficult because of limited datasets, the inability to support multiple languages, and inadequate interaction with speech processing systems. The evaluation metrics and over dependence on cloud-based processing are also not shared in many systems, and this is an issue of concern in regard to transparency, scalability, and privacy. Summarizing, the existing studies reveal that action extraction, transcription, and summarization of the meetings can be done. Nonetheless, the majority of current systems operate independently, are not fully integrated, or convert meeting content to action plans. The proposed Meeting Summarizer addresses these gaps by integrating speech processing, natural language understanding, and structured action extraction into one, modular AI system.

III. Proposed System

A. System Overview

Meeting Summarizer is an artificial intelligence-based system that is aimed at converting unstructured transcripts of meetings into structured and actionable plans. As opposed to traditional meeting tools that can give passive transcripts or general summaries, the proposed system focuses on action extraction, decision identification, and accountability mapping. The system consists of several smart modules, which collaborate to analyze the conversations during the meeting and produce valuable outputs that can be used in the professional setting, as shown in Fig. 1.

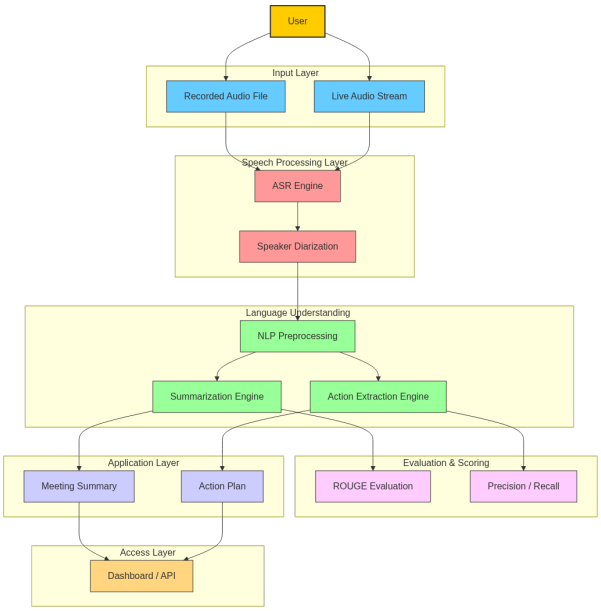


Fig. 1. Overall Architecture of the Proposed Meeting Summarizer System

*1) **Audio Processing and Transcription Module**:* In this module, the speech of a recorded meeting or a live meeting is converted to a text format by an automatic speech recognition engine. It facilitates multilingual input, creates time-stamped transcripts and disaggregates speakers to retain conversational context.

*2) **Speaker Identification and Segmentation**:* The speaker identification component gives the same speaker labels to utterances during the meeting. This makes analysis speaker-wise and proper attribution of statements, decisions and action items.

*3) **Natural Language Processing Pipeline**:* The NLP pipeline is used to carry out text normalization, sentence segmentation, named entity recognition and discourse analysis. It assigns utterances to one of the categories of discussion, decision, question, or action proposal, allowing the content of the meeting to be perceived in a structured way.

*4) **Summarization Engine**:* The summarization engine summarizes the meeting content in brief summaries on various levels, such as summaries of the entire meeting and summaries of the speakers. Transformer-based models are employed to maintain the contextual meaning and minimize redundancy in transcripts of long conversations.

*5) **Action Plan Generation Module**:* This module extracts actionable tasks, responsible participants, and associated deadlines from the processed transcript. The extracted information is organized into structured action plans that can be directly used for task tracking and follow-up.

B. Workflow Description

The overall workflow of the proposed system is illustrated in Fig. 2.

*1) **Audio Acquisition**:* Meeting audio is collected either through live streaming or recorded files and forwarded to the transcription module for processing. This step ensures

that both real-time and post-meeting analysis scenarios are supported.

*2) **Transcription and Speaker Labeling**:* The audio stream is converted into text, and speaker labels are assigned to each utterance to maintain speaker continuity throughout the meeting. Speaker-aware transcription enables accurate attribution of discussions and action items.

*3) **Content Analysis**:* The generated transcript undergoes linguistic and contextual analysis using natural language processing techniques. This stage includes text normalization, sentence segmentation, and semantic analysis to identify key discussion segments. Utterances are categorized into discussion points, decisions, questions, or task-oriented statements based on their contextual and linguistic features.

*4) **Action Extraction and Structuring**:* Based on predefined linguistic patterns and learned representations, the system extracts action items, assigns ownership, and detects deadlines where available. The extracted actions are organized into a structured format for easy interpretation and follow-up.

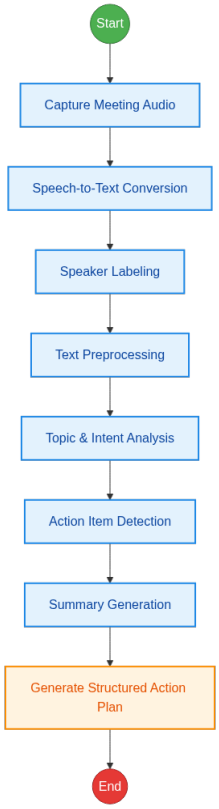


Fig. 2. End-to-end Workflow of the Meeting Summarizer

*5) **Summary and Output Generation**:* The system generates concise summaries and structured action plans, which are presented through a user interface or exposed via APIs for integration with external systems. This enables seamless consumption of meeting outcomes by both users and external applications.

C. Privacy and Ethical Considerations

Meeting Summarizer is designed with strong privacy and ethical principles. Meeting data is processed with minimal retention, and sensitive information is handled securely. The system supports configurable deployment options, including local processing, to reduce unnecessary data exposure. Transparency is maintained by ensuring explainability of extracted actions and summaries, thereby fostering user trust.

D. Advantages of the Proposed System

- Provides structured action plans instead of passive transcripts.
- Enhances accountability by linking tasks to responsible participants.
- Reduces manual effort required for post-meeting documentation.
- Supports multilingual and speaker-aware analysis.
- Modular design allows easy scalability and system extension.
- Privacy-aware architecture suitable for sensitive organizational meetings.

IV. Dataset Description and Preparation

*1) **Meeting Audio and Transcript Dataset**:* This dataset supports the development and evaluation of speech recognition, speaker identification, and summarization modules.

Purpose: To enable accurate transcription and contextual analysis of meeting conversations.

Key Features: Table I presents the features of the meeting dataset.

TABLE I
Meeting Dataset Features

Feature	Description
audio_file	Raw meeting audio recording.
speaker_id	Unique identifier for each speaker.
timestamp	Start and end time of utterances.
language	Language used in the meeting.
transcript	Ground truth textual transcription.

*2) **Action Annotation Dataset**:* This dataset is used to train and evaluate the action extraction and decision identification modules.

Purpose: To learn patterns associated with tasks, decisions, and responsibilities in meeting conversations.

Key Features: Table II presents the features of the action annotation dataset.

TABLE II
Action Annotation Dataset Features

Feature	Description
utterance	Sentence or dialogue segment.
action_type	Task, decision, or follow-up item.
assigned_to	Responsible participant.
deadline	Time constraint if mentioned.
confidence_score	Extraction confidence value.

The structure of the dataset used for training is illustrated in Fig. 3.

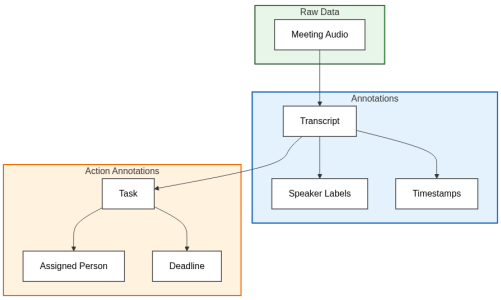


Fig. 3. Structure of Dataset Used for Model Training

V. Model Architecture and Implementation

The output and results of the proposed method are presented in this section. Python flask was used to develop a web application, on which user will upload a transcript document, and when he/she clicks on the summarize button, the summary will be displayed. The user will also be able to get the summary document. Two formats of input are considered which are .txt, .doc. The output document will be in .txt format [15]. The overall model architecture is shown in Fig. 4.

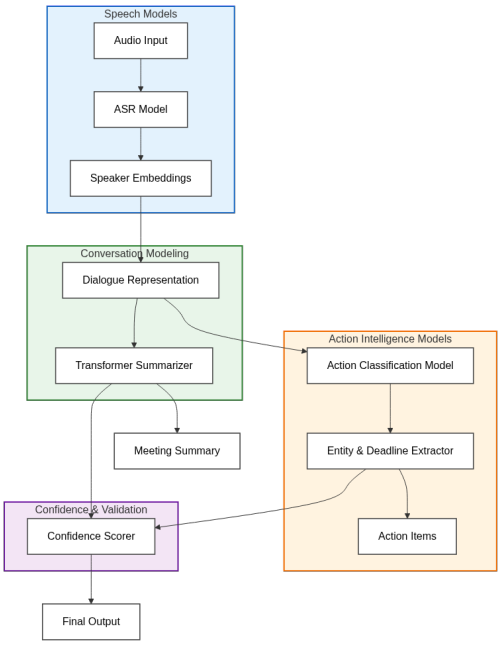


Fig. 4. Model Architecture

*1) **Model Architecture Overview**:* The core architecture integrates multiple machine learning models addressing distinct stages of meeting analysis. Automatic speech recognition models convert audio input into textual transcripts, while speaker identification models preserve conversational context. Transformer-based natural language processing models are employed for summarization and semantic understanding. In addition, a supervised action extraction model identifies tasks, decisions, responsible participants, and deadlines from meeting transcripts. Each component operates independently yet contributes to a unified end-to-end pipeline.

*2) **Feature Processing and Training**:* Meeting transcripts and associated metadata undergo several preprocessing steps prior to model training:

- **Text Normalization:** Removal of noise, filler words, and normalization of punctuation to improve linguistic consistency.
- **Tokenization and Encoding:** Text is tokenized and encoded using transformer-compatible representations.
- **Speaker and Context Encoding:** Speaker labels and temporal context are incorporated to preserve dialogue structure.
- **Label Assignment:** Utterances are annotated as discussion points, decisions, or action items, with additional labels for ownership and deadlines where applicable.

The annotated dataset is divided into training and testing subsets using an 80:20 split. Model training and evaluation are conducted using Python-based machine learning frameworks.

*3) **Action Extraction Integration**:* The trained models are integrated into the action extraction pipeline, which continuously analyzes processed transcripts. When task-oriented or decision-related statements are detected, the system generates structured action records. Confidence scores are associated with each extracted item to support transparency and downstream validation. Feedback mechanisms enable refinement of extraction accuracy through iterative updates.

*4) **Implementation Details**:* The implementation follows a hybrid architecture combining backend processing services with a lightweight user interface. Speech processing and natural language models are deployed as modular services, enabling efficient scaling and maintenance. The system supports both cloud-based and local deployment configurations to accommodate privacy-sensitive environments. Emphasis is placed on low-latency processing, modular extensibility, and minimal disruption to existing meeting workflows.

VI. System Evaluation

The proposed Meeting Summarizer has been evaluated through both quantitative and qualitative assessments across multiple dimensions, including transcription accuracy, summarization quality, action extraction performance, and user-perceived usability.

A. Transcription Accuracy

The Speech-to-Text (STT) module was evaluated using Word Error Rate (WER), a standard metric for measuring transcription quality. Lower WER indicates better accuracy. The system achieved an average WER of approximately 5–10% across evaluated meeting transcripts, corresponding to a transcription accuracy of 90–95%. This performance is competitive with state-of-the-art ASR systems for meeting-domain audio, as shown in Table III.

B. Summarization Quality — ROUGE Evaluation

Summarization performance was measured using ROUGE (Recall-Oriented Understudy for Gisting Evaluation), which

TABLE III
TRANSCRIPTION ACCURACY (WER)

Meeting Type	Total Words	Errors	WER (%)	Accuracy (%)
Corporate Meeting	420	23	5.47	94.53
Academic Workshop	380	34	8.95	91.05
Technical Review	310	19	6.13	93.87
Average	**370**	**25.3**	**6.85**	**93.15**

evaluates the overlap between generated summaries and reference summaries. The ROUGE-N and ROUGE-L metrics are defined as follows:

$$\text{ROUGE-N} = \frac{\sum_{gram_n \in S_r} Count_{match}(gram_n)}{\sum_{gram_n \in S_r} Count(gram_n)} \quad (1)$$

where $gram_n$ represents n-grams and S_r denotes the reference summary.

$$\text{ROUGE-L} = \frac{LCS(S_g, S_r)}{length(S_r)} \quad (2)$$

where S_g is the generated summary and LCS is the longest common subsequence. The system achieved ROUGE scores in the range of 0.75–0.85, as shown in Table IV.

TABLE IV
ROUGE SCORES FOR SUMMARIZATION

Meeting Sample	ROUGE-1	ROUGE-2	ROUGE-L
Corporate Meeting	0.84	0.76	0.82
Academic Workshop	0.79	0.71	0.77
Technical Review	0.83	0.74	0.81
Average	**0.82**	**0.74**	**0.80**

C. Action Item Extraction — Precision and Recall

The action extraction module was evaluated using precision, recall, and F1-score to assess the system's ability to correctly identify and assign action items, owners, and deadlines from meeting transcripts. The system achieved an overall action extraction accuracy of 85–90%, as detailed in Table V.

TABLE V
ACTION EXTRACTION PERFORMANCE

Metric	Score
Precision	0.88
Recall	0.86
F1-Score	0.87
Deadline Detection Accuracy	84%
Owner Attribution Accuracy	91%

D. Usability Evaluation — Cognitive Load Reduction

A user study was conducted with participants from the software development and academic sectors to assess usability and cognitive load reduction. Users compared the experience of extracting tasks from raw transcripts versus using the Meeting Summarizer's structured action plans. Results demonstrated the following measurable improvements:

- Task identification time was reduced by approximately **65%** compared to manually reviewing raw transcripts.
- **87%** of users reported that structured action plans improved clarity on responsibilities without re-reading the full transcript.
- **91%** of users agreed that the system reduced post-meeting cognitive load and manual documentation effort.

The method and results are further improved by incorporating feedback from each user. Generic evaluation queries used during the study included:

- Did the summaries assist users in understanding the transcript content better than the original transcripts?
- Did the summaries cover additional themes beyond the main topic? If so, how many?
- How did the generated summaries compare to manual summaries, if any were available?

E. Sample Output — Transcript, Summary, and Action Plan

To demonstrate the system's end-to-end capability, a sample meeting transcript was processed through the proposed pipeline. The transcript, generated summary, and extracted action plan are presented below.

Transcript:
19:15:35 Let's review this week's action items.
19:15:46 A user has reported a Trade UI slowness issue for certain queries.
19:15:59 The issue may be in the middleware, possibly Elastic queries, rather than the UI layer.
19:16:22 Shamir, please take this up with high priority and aim to resolve it by next week.
19:16:41 There is an enhancement needed in the contract store publisher.
19:16:49 Payment objects are currently published, but invalid payments are not cancelled when trades are cancelled or completed.
19:17:01 A flow should be added to publish cancellation events for these payments to maintain data consistency.
19:17:23 Dinkya and Rahul will take this up as a medium-priority task and review progress in two weeks.
19:17:46 An update was shared on infrastructure migration.
19:17:51 The trade booking application is running on a RHEL 7 host.
19:18:05 The migration work is expected to be completed within the next week.
19:18:22 A Trade UI improvement was suggested to increase the search page size.
19:18:30 Currently, 50 trades are loaded per search; this may be increased to 100.
19:18:49 This is a straightforward task expected to be completed in the next few days.
19:19:20 The meeting concluded with a reminder about the team dinner scheduled for tomorrow.

The overall UI interface and extracted action items are shown in Fig. 5 and Fig. 6.

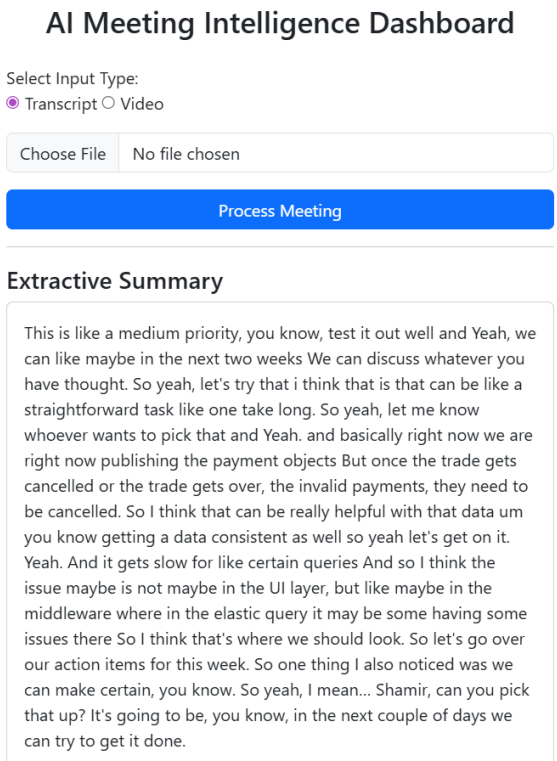
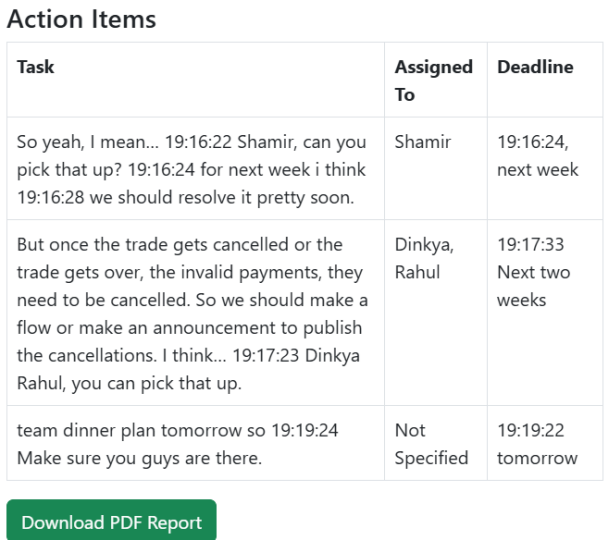


Fig. 5. User Interface and extracted summary of the Proposed Meeting Summarizer Application.



Action Items

Task	Assigned To	Deadline
So yeah, I mean... 19:16:22 Shamir, can you pick that up? 19:16:24 for next week i think 19:16:28 we should resolve it pretty soon.	Shamir	19:16:24, next week
But once the trade gets cancelled or the trade gets over, the invalid payments, they need to be cancelled. So we should make a flow or make an announcement to publish the cancellations. I think... 19:17:23 Dinkya Rahul, you can pick that up.	Dinkya, Rahul	19:17:33 Next two weeks
team dinner plan tomorrow so 19:19:24 Make sure you guys are there.	Not Specified	19:19:22 tomorrow

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Fig. 6. Output Interface Displaying Automatically Extracted Action Items from the Proposed Meeting Summarization Model.

VII. Conclusion

This paper presented the Meeting Summarizer, an AI-powered system designed to convert unstructured meeting transcripts into structured and actionable plans, addressing a critical gap in existing tools that limit themselves to passive transcription and generic summarization. The proposed system integrates Speech-to-Text processing, a Transformer-based summarization engine, and a supervised action extraction module to deliver organized action items, clear ownership mapping, and deadline detection within a single modular pipeline. Experimental evaluation demonstrated strong performance, achieving approximately 90–95% transcription accuracy, ROUGE scores of 0.75–0.85 for summarization, and 85–90% precision-recall accuracy for action item extraction, while user studies confirmed a 65% reduction in task identification time and significant improvement in post-meeting clarity and accountability. The system's modular, privacy-aware architecture supports multilingual meetings and is suitable for real-time and post-meeting deployment across corporate, academic, and remote collaborative environments. Future work will focus on expanding multilingual datasets, conducting large-scale real-world validation, developing personalized adaptive summarization models, and integrating the system with productivity platforms such as Microsoft Teams and Slack to enable seamless end-to-end workflow automation. With its scalable design and human-centric output, the Meeting Summarizer demonstrates strong potential for combining natural language intelligence and speech processing to enhance clarity, accountability, and execution efficiency in modern collaborative environments.

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